

Learning under Endogenous Feedback: Finite-Response Gradients and Joint Stability

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Abstract

Learning systems with endogenous feedback change the environment or population from which subsequent losses are evaluated. Stationary-response methods account for this effect by differentiating through the eventual environment induced by a frozen model. We study the finite-response regime between frozen-environment learning and full stationary-response optimization. A K -step response correction introduces geometric truncation bias but limits propagation of noisy short-run response estimates. We prove a stochastic response-depth theorem that separates these effects in simultaneous high-probability and RMS bounds and yields an explicit approximate preferred horizon $K^*(q, \sigma, n)$ in the slow-response regime. We then derive a finite-sample uncertainty theorem for the actual rolling Stateful PerfGD Jacobian estimator. The bound allows adaptive, temporally dependent regressors, exposes weak excitation through the observed pseudoinverse geometry, and accounts explicitly for measurement error in the lagged population state. A covariance-aware refinement keeps the exact shared-reference differencing structure instead of bounding every difference separately; in the noisy ToyLinear controls it reduces the median certified Jacobian radius by about a factor of 4.6. A further direction-aware ellipsoid propagates anisotropic Jacobian uncertainty through the realized response recursion; it improves the hardest very-slow/noisy selector but is not uniformly better, revealing a remaining distinction between sensitivity certification and online optimization loss. The resulting confidence-bound selector recovers the best evaluated finite depth in three of four regimes and errs conservatively in the fourth. We also derive the full learner–environment Jacobian and exhibit an explicit -1 stability crossing with a computable period-two branch. Matched-budget Stateful PerfGD experiments show that explicit Tikhonov and spectral regularization recover much of the robustness lost by an unregularized stationary inverse, so finite response is not the unique remedy. No method dominates uniformly: stationary regularization is best in one slow noisy regime, whereas shallow finite horizons remain best in very-slow and nonlinear noisy tests. An adaptive trust rule then reduces response depth when local response estimates become unreliable. Finite response is therefore a response-horizon regularizer with a distinct dynamical interpretation, not a universally superior replacement for stationary performative optimization.

Keywords: performative prediction, endogenous feedback, stateful learning, finite-response gradient, implicit differentiation, response regularization, joint stability, dynamic transition.

1 Introduction

Learning under endogenous feedback differs from ordinary empirical risk minimization because the deployed model helps determine the state or distribution from which future losses are evaluated. This occurs in performative prediction, repeated deployment, recommender and policy systems, controlled Markov environments, and other settings in which decisions alter subsequent data. Performative prediction formalizes model-dependent distributions and performative stability [1];

PerfGD and Stateful PerfGD differentiate long-run risk through model-induced responses [2, 3]; related work studies stateful populations, neural predictors, Markovian response, plug-in response models, arbitrary distribution maps, and performative control [4–9].

The stationary-response derivative is therefore not new. The question studied here is different: *what is gained by retaining a finite response horizon and analyzing the learner and environment as a simultaneous dynamical system?* This question matters when a stationary response is expensive to estimate, when derivative estimates are noisy, or when online instability occurs before the environment reaches its frozen-parameter equilibrium.

A related methodological motivation comes from two recent economic-dynamics papers by Melinda L. Wang. Wang [10] formulates macroscopic adjustment as potential descent and classifies equilibrium transitions as structural parameters vary, while Wang [11] couples a Hamiltonian optimization block to a dissipative adaptive-learning block whose target is generated by realized aggregate outcomes. These papers are methodological antecedents rather than machine-learning prior art: the present work replaces the economic state by an explicit endogenous population/data state, studies a discrete response map, and asks how a deliberately finite response horizon changes statistical sensitivity and joint online stability.

Let $x \in \mathbb{R}^m$ denote an environment or population state and $\theta \in \mathbb{R}^d$ the learner. In continuous time we use

$$\dot{x} = A(x, \theta), \quad \dot{\theta} = -\alpha M(x, \theta) \nabla_{\theta} F(x, \theta), \quad M \succeq 0, \quad (1)$$

and in discrete time the environment responds through

$$x_{k+1} = \Phi(x_k, \theta_k). \quad (2)$$

When $x^*(\theta) = \Phi(x^*(\theta), \theta)$ exists, the total derivative of $\bar{F}(\theta) = F(x^*(\theta), \theta)$ contains the standard fixed-point sensitivity

$$\nabla \bar{F} = F_{\theta} + \Phi_{\theta}^T (I - \Phi_x)^{-T} F_x. \quad (3)$$

We use (3) only as a reference endpoint. The finite-response direction truncates propagation after K response steps, creating a controllable bias–sensitivity tradeoff.

Contributions. The paper makes three main contributions.

- (i) **Finite response as regularization.** We derive geometric truncation error and compare response-estimation error for the stationary inverse and a finite response depth. A stochastic response-depth theorem then separates geometric truncation bias from Jacobian-estimation error, gives simultaneous high-probability and RMS bounds, and yields an approximate preferred horizon $K^*(q, \sigma, n)$ in the slow-response regime. We next derive a finite-sample uncertainty radius for the rolling finite-difference regression used by Stateful PerfGD. The radius is valid for adaptive/dependent trajectory regressors, penalizes poor excitation through the observed pseudoinverse geometry, and treats the noisy lagged state as an errors-in-variables term. A covariance-aware version represents all repeated-reference differences through incidence matrices of the unique population-sampling errors and self-normalizes that correlated noise by the observed rolling design. This yields a confidence-bound depth selector without textbook OLS standard-error assumptions. We further replace the resulting scalar Jacobian radius by a direction-aware covariance ellipsoid and propagate that ellipsoid through the realized finite-response recursion. The ellipsoidal bound is exact at the level of sensitivity-error recursion and can be materially tighter in dynamically relevant directions, although its corresponding online selector is not uniformly superior in the experiments.

- (ii) **Joint online stability and transitions.** We analyze the spectrum of the full learner–environment Jacobian, not only the learner Hessian. A nonlinear coupled example exhibits an explicit -1 crossing and a computable period-two branch. This complements prior work showing that performative and multi-agent learning can undergo instability and chaos [12] and work on sensitivity propagation in performative control [9].
- (iii) **Matched response-estimation experiments and adaptive horizon.** We reproduce the rolling-Jacobian estimator used by Stateful PerfGD on its ToyLinear environment and compare full stationary sensitivity with finite-depth propagation under identical observations and deployment counts. Clean estimates favor Stateful PerfGD; slow noisy estimates favor shallow response horizons. A nonlinear stateful benchmark reproduces the reversal. We then introduce an adaptive trust rule that selects response depth from the current correction magnitude and, for estimated response maps, the regression residual.

The claim is intentionally narrower than proposing a universally superior performative optimizer. Finite response is studied as a response-horizon regularizer and as a dynamical object between frozen-environment learning and full stationary-response optimization.

2 Related Work and Novelty Boundary

PerfGD differentiates performative risk through a model-induced distribution [2]. Stateful PerfGD is closer to the present setting: a population state reacts gradually and its stationary sensitivity is estimated from rolling response observations [3]. Brown et al. study stateful population response [4]; Mofakhami et al. analyze performative prediction with neural networks [5]; Liu, Li, and Wai treat Markovian data with a two-timescale derivative-free method [6]; Lin and Zrnic study plug-in performative optimization [7]; and Zhong et al. consider arbitrary distribution maps [8]. Recent adaptive-gradient work further broadens the class of performative optimization problems treated by finite-difference response estimation [13]. Statistical estimation from a single dependent trajectory has a separate literature in linear-system identification and self-normalized least squares [14–16], while noisy regressors lead to classical errors-in-variables effects and require uncertainty analyses different from ordinary independent-design OLS [17, 18]. Our rolling-regression theorem below is deliberately narrower: it is a finite-sample perturbation bound tailored to the reference-difference Jacobian estimator used by Stateful PerfGD. Because the noisy lagged state makes the design itself an errors-in-variables object, we do not invoke a predictable-regressor martingale theorem directly; instead we retain the exact shared-reference covariance geometry and normalize it by the observed design.

Fixed-point differentiation and truncated hypergradients are standard in implicit and bilevel optimization [19–21]. Accordingly, neither $(I - \Phi_x)^{-1}\Phi_\theta$ nor Neumann truncation is claimed as new. Our novelty claim concerns their role as a *finite response horizon* in simultaneous learner–environment dynamics, the associated response-estimation regularization, and the way the resulting direction changes the joint stability boundary.

Dynamic effects also have precedents. Piliouras and Yu prove transitions from stability to instability and chaos in multi-agent performative prediction [12]; Cai, Han, and Cao study temporal sensitivity propagation in performative control [9]; and large-learning-rate and game-dynamics work emphasizes spectral effects beyond small-step optimization [22, 23]. The present paper focuses on a different axis: how deliberately limiting the response horizon changes bias, estimated sensitivity, and the spectrum of the simultaneous online update.

3 Coupled Learner–Environment Dynamics

Assume $F \in C^2$, A is locally Lipschitz, and $M = M^T \succeq 0$ is continuous. Equation (1) then defines the coupled evolution locally.

Theorem 3.1 (Potential balance). *Along every classical solution of (1),*

$$\frac{dF}{dt} = C - D, \quad C := \nabla_x F^T A, \quad D := \alpha \nabla_\theta F^T M \nabla_\theta F \geq 0. \quad (4)$$

For frozen x , F is nonincreasing.

Proof. Apply the chain rule and substitute (1). Positive semidefiniteness of M gives $D \geq 0$. $\square$

The sign of C matters. An absolute-value diagnostic obscures whether the environment injects or removes potential. We therefore replace the diagnostic used in the first draft by the signed positive-injection ratio

$$\chi_+(x, \theta) := \frac{[C(x, \theta)]_+}{D(x, \theta)} \quad (5)$$

whenever $D > 0$.

Corollary 3.2 (Exact instantaneous interpretation of χ_+). *Suppose $D > 0$. If $\chi_+ < 1$, then $\dot{F} < 0$. If $C > 0$ and $\chi_+ > 1$, then $\dot{F} > 0$. The threshold $\chi_+ = 1$ is exactly the balance $C = D$.*

This statement is tautological once the balance law is written, but it gives the diagnostic a precise meaning: χ_+ is not a generic “coupling strength”; it is the fraction of instantaneous adaptive dissipation offset by *positive* environmental injection.

3.1 Finite-step diagnostic

Consider an Euler realization

$$\Delta x = hA(x, \theta), \quad \Delta \theta = -h\alpha M(x, \theta)F_\theta(x, \theta), \quad (6)$$

with $h > 0$. Define

$$S(x, \theta) := \|A\|^2 + \alpha^2 \|MF_\theta\|^2. \quad (7)$$

Theorem 3.3 (Finite-step sign certificate). *Assume $\|\nabla^2 F\| \leq L$ on the line segment joining (x, θ) and $(x + \Delta x, \theta + \Delta \theta)$. Then*

$$|F(x + \Delta x, \theta + \Delta \theta) - F(x, \theta) - h(C - D)| \leq \frac{Lh^2}{2}S. \quad (8)$$

If $D > 0$, $0 < \gamma < 1$, and

$$h \leq \frac{2\gamma D}{LS}, \quad (9)$$

then:

(i) if $\chi_+ \leq 1 - \gamma$, the realized Euler step satisfies $F_{k+1} \leq F_k$;

(ii) if $C > 0$ and $\chi_+ \geq 1 + \gamma$, the realized Euler step satisfies $F_{k+1} \geq F_k$.

Proof. Taylor’s theorem with remainder gives (8) because $\|(\Delta x, \Delta \theta)\|^2 = h^2 S$. If $\chi_+ \leq 1 - \gamma$, then $C - D \leq -\gamma D$ (and the conclusion is even stronger when $C \leq 0$). Combining this with (9) makes the positive Taylor remainder no larger than $h\gamma D$. The second statement is analogous. $\square$

Theorem 3.3 is the main reason to retain χ_+ in the revised paper. We do *not* use χ_+ as an automatically optimal mixing weight for an optimizer; the earlier heuristic rule $\gamma_k = \chi_k/(1 + \chi_k)$ has been removed from the headline algorithmic claims.

4 Stationary Response and Prior-Art Hypergradient

Let

$$x_{j+1} = \Phi(x_j, \theta) \quad (10)$$

with θ held fixed.

Assumption 4.1 (Contractive response). For each θ in an open set, $x \mapsto \Phi(x, \theta)$ is C^1 and satisfies $\|\Phi_x\| \leq q < 1$ in the region of interest.

There is then a unique fixed response $x^*(\theta) = \Phi(x^*(\theta), \theta)$.

Proposition 4.2 (Stationary-response derivative; standard). *Under Assumption 4.1,*

$$D_\theta x^* = (I - \Phi_x)^{-1} \Phi_\theta, \quad (11)$$

and for $\bar{F}(\theta) = F(x^*(\theta), \theta)$,

$$\nabla \bar{F} = F_\theta + \Phi_\theta^T (I - \Phi_x)^{-T} F_x. \quad (12)$$

Remark 4.3 (Novelty boundary). Equation (12) is not a new contribution. It is a fixed-point hypergradient and is closely aligned with the stationary-response sensitivities used in PerfGD and Stateful PerfGD [2, 3] and with generic implicit-differentiation methods [19, 20]. We retain it because it provides the reference limit against which finite-response corrections can be measured.

5 Finite-Response Coupling Corrections

Solve the adjoint equation

$$(I - \Phi_x)^T v = F_x \quad (13)$$

by the truncated iteration

$$v_0 = 0, \quad v_{j+1} = F_x + \Phi_x^T v_j, \quad j = 0, \dots, K-1, \quad (14)$$

and define

$$g_K := F_\theta + \Phi_\theta^T v_K. \quad (15)$$

The case $K = 0$ is the frozen-environment gradient; $K \rightarrow \infty$ gives the stationary-response gradient under contraction.

Proposition 5.1 (Geometric truncation error). *Under Assumption 4.1,*

$$\|v - v_K\| \leq \frac{q^K}{1-q} \|F_x\|, \quad (16)$$

and therefore

$$\|\nabla \bar{F} - g_K\| \leq \|\Phi_\theta\| \frac{q^K}{1-q} \|F_x\|. \quad (17)$$

Proof. Use the Neumann series $(I - \Phi_x^T)^{-1} = \sum_{j \geq 0} (\Phi_x^T)^j$ and bound the tail geometrically. $\square$

The same truncation also changes how errors in an estimated response Jacobian are propagated. This matters when the environment reacts slowly, because stationary sensitivity requires inversion near a unit response multiplier.

Proposition 5.2 (Response-estimation error: stationary inverse versus finite depth). *Let D_1, D_2 be the true short-run response derivatives and let $\hat{D}_1 = D_1 + E_1$, $\hat{D}_2 = D_2 + E_2$. Define*

$$S_\infty = (I - D_2)^{-1} D_1, \quad \hat{S}_\infty = (I - \hat{D}_2)^{-1} \hat{D}_1.$$

Whenever the displayed inverses exist,

$$\hat{S}_\infty - S_\infty = (I - \hat{D}_2)^{-1} (E_1 + E_2 S_\infty), \quad (18)$$

and therefore

$$\|\hat{S}_\infty - S_\infty\| \leq \|(I - \hat{D}_2)^{-1}\| (\|E_1\| + \|E_2\| \|S_\infty\|). \quad (19)$$

For the depth- K sensitivities

$$S_K = \sum_{j=0}^{K-1} D_2^j D_1, \quad \hat{S}_K = \sum_{j=0}^{K-1} \hat{D}_2^j \hat{D}_1,$$

if $\|D_2\|, \|\hat{D}_2\| \leq \bar{q}$, then

$$\|\hat{S}_K - S_K\| \leq \left(\sum_{j=0}^{K-1} \bar{q}^j \right) \|E_1\| + \|D_1\| \|E_2\| \sum_{j=1}^{K-1} j \bar{q}^{j-1}. \quad (20)$$

Thus, for fixed K , the finite-depth error bound remains finite as $\bar{q} \uparrow 1$, whereas the stationary inverse can amplify derivative-estimation error through the resolvent norm in (19).

Proof. For the stationary sensitivity, multiply $\hat{S}_\infty - S_\infty$ by $I - \hat{D}_2$ and use $(I - D_2)S_\infty = D_1$ to obtain (18). For finite depth,

$$\hat{S}_K - S_K = \sum_{j=0}^{K-1} \hat{D}_2^j E_1 + \sum_{j=0}^{K-1} (\hat{D}_2^j - D_2^j) D_1.$$

The telescoping identity

$$\hat{D}_2^j - D_2^j = \sum_{\ell=0}^{j-1} \hat{D}_2^\ell E_2 D_2^{j-1-\ell}$$

gives $\|\hat{D}_2^j - D_2^j\| \leq j \bar{q}^{j-1} \|E_2\|$, yielding (20). □

Remark 5.3 (Slow-response regularization). For the ToyLinear response used below,

$$m(\theta, \mu) = \delta(A_T \theta + b) + (1 - \delta)\mu,$$

so $D_1 = \delta A_T$, $D_2 = (1 - \delta)I$, and $S_\infty = A_T$. The true stationary inverse gain is

$$\|(I - D_2)^{-1}\| = \frac{1}{\delta}.$$

Hence exact stationary sensitivity is unbiased when D_1, D_2 are known, but noisy estimation of the short-run response can be strongly magnified as $\delta \downarrow 0$. A fixed finite depth introduces truncation bias but avoids resolving the full stationary pole. Proposition 5.2 formalizes this bias-estimation-error tradeoff.

Theorem 5.4 (Stochastic response-depth error and oracle horizon). *Let D_1, D_2 be the true short-run response derivatives with $\|D_2\| \leq q < 1$, and let $\widehat{D}_i = D_i + E_i$ be estimates constructed from n response observations. Suppose that for $i = 1, 2$ and every $\delta_0 \in (0, 1)$,*

$$\Pr(\|E_i\| > \varepsilon_i(n, \delta_0)) \leq \frac{\delta_0}{2}, \quad \varepsilon_i(n, \delta_0) := \sigma_i \sqrt{\frac{2 \log(4/\delta_0)}{n}}. \quad (21)$$

Set $\bar{q}_n := q + \varepsilon_2(n, \delta_0)$. Define

$$A_K(r) := \sum_{j=0}^{K-1} r^j = \frac{1-r^K}{1-r}, \quad B_K(r) := \sum_{j=1}^{K-1} j r^{j-1} = \frac{1-Kr^{K-1}+(K-1)r^K}{(1-r)^2}, \quad (22)$$

with $B_1(r) = 0$, and set $C_0 := \|D_1\|/(1-q)$. Then with probability at least $1 - \delta_0$, simultaneously for every $K \geq 1$,

$$\left\| \widehat{S}_K - S_\infty \right\| \leq C_0 q^K + A_K(\bar{q}_n) \varepsilon_1(n, \delta_0) + \|D_1\| B_K(\bar{q}_n) \varepsilon_2(n, \delta_0). \quad (23)$$

Consequently, for any finite candidate set $\mathcal{K} \subset \{1, 2, \dots\}$, the oracle response depth

$$K_{n, \delta_0}^* \in \arg \min_{K \in \mathcal{K}} \{C_0 q^K + A_K(\bar{q}_n) \varepsilon_1(n, \delta_0) + \|D_1\| B_K(\bar{q}_n) \varepsilon_2(n, \delta_0)\} \quad (24)$$

minimizes this simultaneous high-probability upper bound over the admissible horizons.

Proof. Decompose

$$\widehat{S}_K - S_\infty = (\widehat{S}_K - S_K) + (S_K - S_\infty).$$

Since $\|D_2\| \leq q < 1$, the deterministic truncation tail satisfies

$$\|S_K - S_\infty\| \leq \sum_{j=K}^{\infty} q^j \|D_1\| = C_0 q^K.$$

By a union bound, with probability at least $1 - \delta_0$ both $\|E_i\| \leq \varepsilon_i(n, \delta_0)$ hold. On that event, $\|\widehat{D}_2\| \leq q + \varepsilon_2 = \bar{q}_n$. Proposition 5.2 applies for finite K even if the estimated norm exceeds one, so it gives

$$\left\| \widehat{S}_K - S_K \right\| \leq A_K(\bar{q}_n) \varepsilon_1 + \|D_1\| B_K(\bar{q}_n) \varepsilon_2.$$

Combining the two bounds proves (23). The definition (24) then gives the final statement. $\square$

Corollary 5.5 (RMS response-depth bound). *Suppose instead that*

$$\left(\mathbb{E} \|E_i\|^2 \right)^{1/2} \leq \frac{\sigma_i}{\sqrt{n}}, \quad i = 1, 2,$$

and that $\|\widehat{D}_2\| \leq \bar{q} < 1$ almost surely. Then

$$\left(\mathbb{E} \left\| \widehat{S}_K - S_\infty \right\|^2 \right)^{1/2} \leq C_0 q^K + \frac{\sigma_1}{\sqrt{n}} A_K(\bar{q}) + \frac{\|D_1\| \sigma_2}{\sqrt{n}} B_K(\bar{q}). \quad (25)$$

Proof. Apply the deterministic decomposition in the proof of Theorem 5.4, then use Minkowski's inequality in L^2 together with Proposition 5.2. $\square$

Corollary 5.6 (Slow-response approximation for the preferred depth). *Let $\lambda = -\log q$. In a slow-response regime with $\bar{q} \approx q$, $\lambda K \lesssim 1$, and a candidate horizon below the mixing time,*

$$A_K(q) \approx K, \quad B_K(q) \approx \frac{K(K-1)}{2}, \quad q^K = e^{-\lambda K}.$$

The RMS bound is therefore approximated by

$$U_n(K) \approx C_0 e^{-\lambda K} + \frac{1}{\sqrt{n}} \left(\sigma_1 K + \frac{\|D_1\| \sigma_2}{2} K^2 \right). \quad (26)$$

Treating K as continuous, an interior minimizer approximately solves

$$C_0 \lambda e^{-\lambda K} \approx \frac{\sigma_1 + \|D_1\| \sigma_2 K}{\sqrt{n}}. \quad (27)$$

If direct-response derivative noise dominates,

$$K^* \approx \frac{1}{\lambda} \left[\log \left(\frac{C_0 \lambda \sqrt{n}}{\sigma_1} \right) \right]_+, \quad (28)$$

whereas if state-Jacobian noise dominates,

$$K^* \approx \frac{1}{\lambda} W \left(\frac{C_0 \lambda^2 \sqrt{n}}{\|D_1\| \sigma_2} \right), \quad (29)$$

with W the principal Lambert W function. The integer depth is obtained by rounding to the nearest admissible candidate.

Remark 5.7 (Interpretation and data-driven use). Equations (23)–(29) make the bias–variance interpretation quantitative. Larger n supports a deeper response horizon; larger Jacobian-estimation noise supports a shallower horizon; and slow response (q near one) makes the tradeoff sharper because short-run derivative errors are repeatedly propagated. The theorem is estimator-agnostic: it requires valid radii for $\hat{D}_1$ and $\hat{D}_2$. We now derive such a radius for the rolling finite-difference estimator actually used by Stateful PerfGD, rather than importing independent-regressor OLS standard errors.

5.1 Dependent rolling-regression uncertainty and a certified response horizon

Stateful PerfGD estimates the local Jacobian from a short segment of a single optimization trajectory. The regressors are therefore adaptive and temporally dependent, and the lagged population state entering the regressor is itself estimated from samples. These features fall outside the textbook independent-design OLS calculation. They are also precisely the features that matter here: persistent trajectories reduce directional excitation, while population-estimation error enters both the response and the lagged-state covariate. Related dependent-trajectory and errors-in-variables problems are studied more generally in [14, 17, 18]. The following finite-sample bound is instead tailored directly to Algorithm 2 of Stateful PerfGD [3].

Let

$$\psi_s = (\theta_s, \mu_{s-1}) \in \mathbb{R}^{2d}, \quad \hat{\psi}_s = (\theta_s, \hat{\mu}_{s-1}), \quad \xi_s := \hat{\mu}_s - \mu_s.$$

At update t , choose m previous response pairs $s_1, \dots, s_m < t$ and form the same reference-difference matrices used by the rolling estimator,

$$\hat{X}_t = \begin{bmatrix} \hat{\psi}_{s_1} - \hat{\psi}_t & \cdots & \hat{\psi}_{s_m} - \hat{\psi}_t \end{bmatrix}, \quad \hat{Y}_t = \begin{bmatrix} \hat{\mu}_{s_1} - \hat{\mu}_t & \cdots & \hat{\mu}_{s_m} - \hat{\mu}_t \end{bmatrix}. \quad (30)$$

The paper-faithful estimate is

$$\widehat{D}_t = \widehat{Y}_t \widehat{X}_t^\dagger = [\widehat{D}_{1,t}, \widehat{D}_{2,t}]. \quad (31)$$

No stochastic independence of the trajectory $\{\psi_s\}$ is imposed below.

Theorem 5.8 (Covariance-aware rolling-regression uncertainty with a noisy lagged state). *Suppose the response map $m : \mathbb{R}^{2d} \rightarrow \mathbb{R}^d$ is C^2 on a convex neighborhood containing the rolling trajectory, and let*

$$D_t = Dm(\psi_t) = [D_{1,t}, D_{2,t}], \quad \|D_{2,t}\| \leq q_0 < 1, \quad \|D^2 m\| \leq L_m.$$

Assume the observed finite-difference design $\widehat{X}_t$ in (30) has full row rank. Let $\mathcal{U}_t$ be the set of distinct population-estimation times appearing in the response differences $\xi_{s_j} - \xi_t$ and lagged-state differences $\xi_{s_j-1} - \xi_{t-1}$. Enumerate $\mathcal{U}_t = \{u_1, \dots, u_R\}$ and define

$$\Xi_t = [\xi_{u_1} \quad \dots \quad \xi_{u_R}] \in \mathbb{R}^{d \times R}.$$

Let $B_{Y,t}, B_{Z,t} \in \mathbb{R}^{R \times m}$ be the incidence matrices satisfying

$$\Xi_t B_{Y,t} = [\xi_{s_1} - \xi_t \quad \dots \quad \xi_{s_m} - \xi_t], \quad \Xi_t B_{Z,t} = [\xi_{s_1-1} - \xi_{t-1} \quad \dots \quad \xi_{s_m-1} - \xi_{t-1}]. \quad (32)$$

For a consecutive rolling window, for example,

$$B_{Y,t}^T B_{Y,t} = B_{Z,t}^T B_{Z,t} = I_m + \mathbf{1}\mathbf{1}^T,$$

which is exactly the covariance pattern induced by reusing one reference observation in every column; temporal overlap between the response and lagged-state differences is retained through the common matrix Ξ_t .

Write

$$\kappa_t := \sigma_{\min}(\widehat{X}_t) > 0, \quad (33)$$

and define the observable design-normalized shared-reference factor

$$\mathcal{G}_t(q_0) := \|B_{Y,t} \widehat{X}_t^\dagger\| + q_0 \|B_{Z,t} \widehat{X}_t^\dagger\|. \quad (34)$$

On any event for which

$$\max_{r \in \mathcal{U}_t} \|\xi_r\| \leq \varepsilon_\mu, \quad \|\Xi_t\| \leq a_t, \quad (35)$$

define the observable curvature allowance

$$\mathcal{R}_t(\varepsilon_\mu) := \frac{L_m}{2} \left[\sum_{j=1}^m \left(\|\widehat{\psi}_{s_j} - \widehat{\psi}_t\| + 2\varepsilon_\mu \right)^4 \right]^{1/2}. \quad (36)$$

Then

$$\boxed{\|\widehat{D}_t - D_t\| \leq r_t^{\text{cov}}(\varepsilon_\mu, a_t) := \frac{\mathcal{R}_t(\varepsilon_\mu)}{\kappa_t} + a_t \mathcal{G}_t(q_0)}. \quad (37)$$

Consequently

$$\|\widehat{D}_{i,t} - D_{i,t}\| \leq r_t^{\text{cov}}(\varepsilon_\mu, a_t), \quad i = 1, 2.$$

The result is pathwise on (35): the trajectory, $\widehat{X}_t$, and hence $\widehat{X}_t^\dagger$ may depend arbitrarily on earlier population-sampling errors. No independence between the realized pseudoinverse and Ξ_t is used in the inequality.

Proof. Taylor expansion about ψ_t gives

$$\mu_{s_j} - \mu_t = D_t(\psi_{s_j} - \psi_t) + \rho_j, \quad \|\rho_j\| \leq \frac{L_m}{2} \|\psi_{s_j} - \psi_t\|^2.$$

Because only the population-state block of ψ is measured with error,

$$\widehat{\psi}_{s_j} - \widehat{\psi}_t = (\psi_{s_j} - \psi_t) + (0, \xi_{s_j-1} - \xi_{t-1}).$$

Therefore, after collecting columns,

$$\widehat{Y}_t - D_t \widehat{X}_t = R_t + \Xi_t B_{Y,t} - D_{2,t} \Xi_t B_{Z,t}. \quad (38)$$

Since $\widehat{X}_t$ has full row rank,

$$\widehat{D}_t - D_t = R_t \widehat{X}_t^\dagger + \Xi_t B_{Y,t} \widehat{X}_t^\dagger - D_{2,t} \Xi_t B_{Z,t} \widehat{X}_t^\dagger.$$

On (35),

$$\|R_t\| \leq \mathcal{R}_t(\varepsilon_\mu), \quad \left\| R_t \widehat{X}_t^\dagger \right\| \leq \frac{\mathcal{R}_t(\varepsilon_\mu)}{\kappa_t}.$$

For the shared-reference sampling noise, use the same base matrix Ξ_t in both terms rather than bounding the m differences separately:

$$\begin{aligned} & \left\| \Xi_t B_{Y,t} \widehat{X}_t^\dagger - D_{2,t} \Xi_t B_{Z,t} \widehat{X}_t^\dagger \right\| \\ & \leq \|\Xi_t\| \left(\left\| B_{Y,t} \widehat{X}_t^\dagger \right\| + \left\| D_{2,t} \right\| \left\| B_{Z,t} \widehat{X}_t^\dagger \right\| \right) \\ & \leq a_t \mathcal{G}_t(q_0). \end{aligned}$$

This proves (37). The argument is pointwise for every realization satisfying (35), so the fact that $\widehat{X}_t^\dagger$ is data dependent creates no additional conditioning requirement. Restricting columns yields the blockwise bounds. $\square$

Corollary 5.9 (Gaussian shared-reference confidence radius). *Suppose the distinct population-estimation errors in $\mathcal{U}_t$ are generated from fresh samples and satisfy*

$$\xi_r \sim N(0, \sigma_\mu^2 I_d), \quad r \in \mathcal{U}_t,$$

independently across deployments. Let $R = |\mathcal{U}_t|$. Then

$$\Pr \left\{ \|\Xi_t\| \leq a_t(\delta) := \sigma_\mu \left[\sqrt{d} + \sqrt{R} + \sqrt{2 \log(1/\delta)} \right] \right\} \geq 1 - \delta. \quad (39)$$

If $L_m = 0$, substitution of $a_t(\delta)$ into (37) gives directly the pointwise $(1 - \delta)$ radius

$$\left\| \widehat{D}_t - D_t \right\| \leq a_t(\delta) \mathcal{G}_t(q_0). \quad (40)$$

For $L_m > 0$, combine (39) with the simultaneous vector bound

$$\varepsilon_{\mu,t}(\delta) = \sigma_\mu \left[\sqrt{d} + \sqrt{2 \log \frac{R}{\delta}} \right]$$

using a split of the desired failure probability between the two events. If sampled means satisfy $\sigma_\mu \leq v/\sqrt{n_{\min}}$, the same formulas hold with σ_μ replaced by $v/\sqrt{n_{\min}}$.

Proof. For a $d \times R$ matrix with independent $N(0, \sigma_\mu^2)$ entries, the standard Gaussian spectral-norm inequality gives (39). The affine case then follows from Theorem 5.8. For the nonlinear case, apply the usual Gaussian vector-norm bound and a union bound over the R unique sampling errors to control the curvature conversion from observed to latent differences. $\square$

Theorem 5.10 (Direction-aware rolling-Jacobian ellipsoid). *Under the hypotheses and event of Theorem 5.8, write*

$$C_{Y,t} := B_{Y,t} \hat{X}_t^\dagger, \quad C_{Z,t} := B_{Z,t} \hat{X}_t^\dagger, \quad r_{\text{curv},t} := \frac{\mathcal{R}_t(\varepsilon_\mu)}{\kappa_t}.$$

Then the joint Jacobian error $\Delta D_t := \hat{D}_t - D_t \in \mathbb{R}^{d \times 2d}$ satisfies the matrix inequality

$$\Delta D_t^T \Delta D_t \preceq W_t := 2r_{\text{curv},t}^2 I_{2d} + 4a_t^2 (C_{Y,t}^T C_{Y,t} + q_0^2 C_{Z,t}^T C_{Z,t}). \quad (41)$$

If the response is affine on the rolling window, so that $L_m = 0$, the sharper ellipsoid

$$W_t^{\text{aff}} := 2a_t^2 (C_{Y,t}^T C_{Y,t} + q_0^2 C_{Z,t}^T C_{Z,t}) \quad (42)$$

is valid. Thus uncertainty is anisotropic: directions nearly annihilated by the incidence/pseudoinverse geometry receive a small radius even when the global operator-norm certificate is large.

Proof. For any $z \in \mathbb{R}^{2d}$, the sampling-noise part of (38) gives

$$\|\Xi_t C_{Y,t} z - D_{2,t} \Xi_t C_{Z,t} z\| \leq a_t (\|C_{Y,t} z\| + q_0 \|C_{Z,t} z\|).$$

Squaring and using $(u + v)^2 \leq 2u^2 + 2v^2$ yields

$$\|\Xi_t C_{Y,t} z - D_{2,t} \Xi_t C_{Z,t} z\|^2 \leq 2a_t^2 z^T (C_{Y,t}^T C_{Y,t} + q_0^2 C_{Z,t}^T C_{Z,t}) z.$$

The curvature part obeys $\|R_t \hat{X}_t^\dagger z\| \leq r_{\text{curv},t} \|z\|$. A second application of $(u + v)^2 \leq 2u^2 + 2v^2$ gives (41); when $r_{\text{curv},t} = 0$, the first inequality alone gives the sharper affine form (42). $\square$

Proposition 5.11 (Exact direction-aware propagation through response depth). *Let $S_0 = \hat{S}_0 = 0$ and*

$$S_{j+1} = D_1 + D_2 S_j, \quad \hat{S}_{j+1} = \hat{D}_1 + \hat{D}_2 \hat{S}_j.$$

Suppose $\Delta D^T \Delta D \preceq W$ and $\|D_2\| \leq q_0 < 1$. Define

$$A_j := \begin{bmatrix} I_d \\ \hat{S}_j \end{bmatrix}, \quad \omega_j(W) := \sqrt{\lambda_{\max}(A_j^T W A_j)}.$$

Then for every $K \geq 1$,

$$\|\hat{S}_K - S_K\| \leq \sum_{j=0}^{K-1} q_0^{K-1-j} \omega_j(W). \quad (43)$$

If W_{11} is the upper-left $d \times d$ block of W , then

$$\|D_1\| \leq \hat{d}_1^U := \|\hat{D}_1\| + \sqrt{\lambda_{\max}(W_{11})},$$

and therefore

$$\|\hat{S}_K - S_\infty\| \leq U^{\text{ell}}(K) := \frac{\hat{d}_1^U}{1 - q_0} q_0^K + \sum_{j=0}^{K-1} q_0^{K-1-j} \omega_j(W). \quad (44)$$

Proof. With $E_j = \widehat{S}_j - S_j$ and $\Delta D = [\Delta D_1, \Delta D_2]$, direct subtraction gives the exact recursion

$$E_{j+1} = \Delta D_1 + \Delta D_2 \widehat{S}_j + D_2 E_j = \Delta D A_j + D_2 E_j.$$

Hence $\|E_{j+1}\| \leq \omega_j(W) + q_0 \|E_j\|$. Iteration from $E_0 = 0$ proves (43). The bound on D_1 follows by testing the ellipsoid on vectors $(v, 0)$, and the stationary tail satisfies $\|S_\infty - S_K\| \leq \|D_1\| q_0^K / (1 - q_0)$. Combining the two inequalities proves (44). $\square$

Equation (43) is the direction-aware replacement for repeatedly charging the same scalar radius. At depth j , the uncertainty set is queried only in the realized sensitivity direction $A_j = [I, \widehat{S}_j^T]^T$. A certified ellipsoidal selector is therefore

$$\widehat{K}_t^{\text{ell}} \in \arg \min_{K \in \{1, 2, 5, 10\}} U_t^{\text{ell}}(K), \quad (45)$$

with W_t^{aff} in the ToyLinear experiment.

The incidence representation makes precise what the earlier radius discarded. If only the response differences were present, their vectorized covariance would contain the factor $I_m + \mathbf{1}\mathbf{1}^T$, not an identity matrix; the same reference error appears in every column. Equation (34) carries this covariance geometry through the actual observed pseudoinverse. It is therefore a design-self-normalized bound in the geometric sense. It is not an application of the standard predictable-regressor self-normalized martingale theorem: because the lagged population estimate is itself noisy and reused by later trajectory points, the rolling design is an errors-in-variables object. Classical self-normalized least squares and dependent-system identification provide the broader context [14–16]; the pathwise incidence argument above avoids imposing their predictable-design structure where it is not available.

For comparison, the previous coordinatewise treatment used

$$2\varepsilon_\mu \sqrt{m}(1 + q_0) / \kappa_t$$

for the sampling-noise contribution. In the same ToyLinear runs, replacing that term by $a_t \mathcal{G}_t(q_0)$ reduces the median certified radius from 118.5 to 25.6 in the slow/noisy regime and from 159.9 to 34.9 in the very-slow/noisy regime, approximately 4.6-fold in both cases. Thus most of the earlier numerical conservatism came from treating correlated reference differences as unrelated worst-case columns. The remaining gap between $K = 1$ and the empirical $K = 2$ choice in the hardest regime is instead driven by worst-direction propagation of the estimated state Jacobian.

Certified finite-horizon selector. Assume a structural contractivity bound $q_0 < 1$ is available, as in the standard Stateful PerfGD assumptions. Let

$$\widehat{r}_t = r_t^{\text{cov}}(\varepsilon_{\mu,t}(\delta), a_t(\delta)), \quad \widehat{d}_{1,t}^U := \|\widehat{D}_{1,t}\| + \widehat{r}_t, \quad \widehat{q}_t^{\text{prop}} := \max\{q_0, \|\widehat{D}_{2,t}\|\}.$$

Combining Theorem 5.8 with Theorem 5.4 gives the pointwise score

$$\widehat{U}_t^{\text{dep}}(K) = \frac{\widehat{d}_{1,t}^U}{1 - q_0} q_0^K + A_K(\widehat{q}_t^{\text{prop}}) \widehat{r}_t + \widehat{d}_{1,t}^U B_K(\widehat{q}_t^{\text{prop}}) \widehat{r}_t, \quad (46)$$

with

$$\widehat{K}_t \in \arg \min_{K \in \{1, 2, 5, 10\}} \widehat{U}_t^{\text{dep}}(K). \quad (47)$$

Unlike the earlier OLS standard-error plug-in, (46) does not clip an estimated response norm into the contractive region. The true response is controlled by $q_0 < 1$, while a noisy estimate with norm larger than one is allowed to make deeper finite propagation more expensive through A_K and B_K .

Table 1 evaluates (47) on the same ToyLinear controls used previously. ToyLinear is exactly affine, so $L_m = 0$ and the curvature/vector-radius term drops out. We use the single structural bound $q_0 = .995$ in every regime and a pointwise 90% Gaussian matrix radius in Corollary 5.9; neither quantity is retuned by regime. To isolate depth selection, all fixed depths and the selector use the same learning-rate/exploration setting within each regime.

Regime	best K	selected $\hat{K}$	selector gap	median κ_t	median $\hat{r}_t$
fast, clean	10	10	0.01576 ± 0.00007	1.16×10^{-4}	0
slow, clean	10	10	4.8060 ± 0.0020	1.56×10^{-7}	0
slow, noisy	1	1	6.061 ± 0.404	0.0205	25.6
very slow, noisy	2	1	5.961 ± 0.519	0.0152	34.9

Table 1: Dependent-regression confidence-bound selection in ToyLinear. The selector uses the same rolling observations as Stateful PerfGD but replaces textbook OLS standard errors by Theorem 5.8. It recovers the best evaluated depth in three of four regimes and is conservatively one step too shallow in the very-slow/noisy case; there the best fixed- K gap is 5.161 ± 0.222 . The covariance-aware shared-reference radius is about 4.6 times smaller than the previous coordinatewise certified radius in both noisy regimes. The remaining conservatism is therefore not primarily an artifact of the repeated-reference union bound.

The fourth regime remains a useful negative result. The covariance-aware radius is far tighter than the previous coordinatewise certificate, yet the resulting worst-case score still cannot distinguish the empirical advantage of $K = 2$ over $K = 1$. In the two clean affine controls the radius is zero in exact arithmetic and the deepest candidate is selected. This isolates the remaining theoretical conservatism more sharply: it comes from propagating an operator-norm uncertainty set through powers of a noisy $\hat{D}_2$, rather than from mishandling the shared reference. Theorem 5.10 and Proposition 5.11 implement that refinement. Table 2 compares the resulting selector with the scalar covariance-aware certificate on the same twenty held-out ToyLinear seeds.

Regime	scalar gap	ellipsoidal gap	ellip. mean K	ellip. median K	best fixed K
fast, clean	0.01576	0.01576	10.00	10	10
slow, clean	4.8060	4.8060	10.00	10	10
slow, noisy	6.061 ± 0.404	7.125 ± 0.563	4.02	1	1
very slow, noisy	5.961 ± 0.519	5.286 ± 0.310	3.98	1	2

Table 2: Scalar versus direction-aware certified depth selection. The ellipsoidal score materially narrows the gap to the fixed- $K = 2$ oracle in the very-slow/noisy regime (5.161 ± 0.222), but is somewhat too permissive in the ordinary slow/noisy regime. There it selects $K = 10$ on about 30.5% of updates; in the very-slow/noisy regime the corresponding fraction is about 30.2%. Thus anisotropic uncertainty propagation helps in the hardest regime but does not by itself yield a uniformly better online selector.

The experiment sharpens the limitation rather than removing it. The ellipsoidal confidence set contains the true short-run Jacobian on all recorded ToyLinear updates in these held-out runs, yet a bound on sensitivity error is not the same object as a bound on final optimization loss. In particular, occasional deep selections can be dynamically costly even when the local sensitivity certificate is small. A sequentially uniform confidence construction or a joint certificate that couples sensitivity uncertainty to the learner–environment Jacobian is therefore a more plausible next route

than further pointwise tightening alone.

Algorithm 1 K -step Coupling-Aware Potential Descent (CAPD- K)

Require: current (x_k, θ_k) , differentiable response model Φ , learning rate η , depth K

- 1: evaluate $F_x, F_\theta, \Phi_x, \Phi_\theta$ at (x_k, θ_k)
 - 2: $v \leftarrow 0$
 - 3: **for** $j = 0, \dots, K - 1$ **do**
 - 4: $v \leftarrow F_x + \Phi_x^T v$
 - 5: **end for**
 - 6: $g_K \leftarrow F_\theta + \Phi_\theta^T v$
 - 7: $\theta_{k+1} \leftarrow \theta_k - \eta g_K$
 - 8: $x_{k+1} \leftarrow \Phi(x_k, \theta_k)$
-

Algorithm 1 is best interpreted as a controlled response-depth approximation. It is not intended to supplant Stateful PerfGD when an accurate stationary response can be estimated cheaply. Its potential use is when the response model is available locally, stationary estimation is expensive, or the online joint dynamics themselves are part of the object of interest.

6 Joint Online Stability

Consider

$$\begin{aligned} x_{k+1} &= \Phi(x_k, \theta_k), \\ \theta_{k+1} &= \theta_k - \eta G(x_k, \theta_k), \end{aligned} \tag{48}$$

where G may be the frozen gradient, g_K , or the exact online response-corrected direction. At a fixed point (x^*, θ^*) the Jacobian is

$$\mathcal{J}_\eta = \begin{pmatrix} \Phi_x & \Phi_\theta \\ -\eta G_x & I - \eta G_\theta \end{pmatrix}_{(x^*, \theta^*)}. \tag{49}$$

Proposition 6.1 (Joint Schur stability). *If $\rho(\mathcal{J}_\eta) < 1$, the fixed point is locally asymptotically stable. If $\rho(\mathcal{J}_\eta) > 1$, it is unstable.*

The basic criterion is standard. Its relevance is that the stability boundary contains the off-diagonal feedback blocks $\Phi_\theta G_x$; a learner-only Hessian can therefore miss the true online edge of stability.

6.1 A multidimensional stateful quadratic benchmark

We now give a completed benchmark in which the finite- K correction is not collinear with the frozen gradient. Let $x, \theta \in \mathbb{R}^3$ and

$$x_{k+1} = Ax_k + B\theta_k, \tag{50}$$

with

$$A = \begin{pmatrix} .70 & .12 & 0 \\ -.08 & .62 & .10 \\ 0 & -.12 & .55 \end{pmatrix}, \quad B = \begin{pmatrix} .35 & -.10 & .08 \\ .05 & .30 & -.12 \\ -.10 & .08 & .28 \end{pmatrix}.$$

The training potential is

$$F(x, \theta) = \frac{1}{2} \|R\theta - Cx - d\|^2 + \frac{\lambda}{2} \|\theta\|^2, \quad \lambda = .05, \quad (51)$$

where

$$C = \begin{pmatrix} .90 & .25 & -.10 \\ -.20 & .75 & .20 \\ .15 & -.10 & .85 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & .15 & 0 \\ -.10 & 1 & .20 \\ .05 & -.15 & 1 \end{pmatrix}, \quad d = (1, -.6, .4)^T.$$

Since $\rho(A) < 1$, the stationary state is $x^*(\theta) = K_\infty \theta$ with $K_\infty = (I - A)^{-1}B$. Define $Q = R - CK_\infty$. Then

$$\bar{F}(\theta) = \frac{1}{2} \|Q\theta - d\|^2 + \frac{\lambda}{2} \|\theta\|^2 \quad (52)$$

has unique optimum

$$\theta^* = (Q^T Q + \lambda I)^{-1} Q^T d. \quad (53)$$

For finite response depth,

$$P_K = \sum_{j=0}^{K-1} A^j B, \quad D_K = R - CP_K. \quad (54)$$

At a joint fixed point of the online CAPD- K update, $x = K_\infty \theta$ and

$$(D_K^T Q + \lambda I) \theta_K = D_K^T d. \quad (55)$$

Proposition 6.2 (Finite-depth fixed-point consistency). *Assume $\|A\| \leq q < 1$ and $Q^T Q + \lambda I \succ 0$. Then for all sufficiently large K , the matrix in (55) is invertible and*

$$\|\theta_K - \theta^*\| = O(q^K), \quad \bar{F}(\theta_K) - \bar{F}(\theta^*) = O(q^{2K}). \quad (56)$$

Proof. The tail identity

$$K_\infty - P_K = A^K (I - A)^{-1} B \quad (57)$$

gives $\|D_K - Q\| = O(q^K)$. Hence $D_K^T Q + \lambda I$ is an $O(q^K)$ perturbation of $Q^T Q + \lambda I$ and is invertible for large K . Standard inverse perturbation then yields $\|\theta_K - \theta^*\| = O(q^K)$. Since θ^* minimizes the quadratic $\bar{F}$, Taylor expansion around θ^* gives the $O(q^{2K})$ objective gap. $\square$

Table 3 reports the actual fixed points and stability boundaries obtained by direct eigendecomposition of the joint update matrix. The “oracle stationary gradient” uses the exact gradient of $\bar{F}(\theta)$ and updates θ independently of the current lagged x_k ; it is an idealized stationary-response baseline, not a reimplement of the estimator in Stateful PerfGD.

Method	largest stable η	$\bar{F}(\theta_K) - \bar{F}(\theta^*)$	$\ \theta_K - \theta^*\ $
ordinary online ($K = 0$)	1.518	3.8004	12.0218
CAPD-1	2.026	2.3762	9.4249
CAPD-2	2.749	1.2551	6.7419
CAPD-5	3.563	0.07474	1.4471
CAPD-10	3.728	0.000325	0.04976
CAPD-exact online	3.741	0	0
Oracle stationary gradient	5.137	0	0

Table 3: Completed 3D quadratic benchmark. Increasing response depth moves the online fixed point toward the stationary-response optimum and, for this example, enlarges the observed nominal stability interval. The oracle stationary-gradient baseline remains more stable, underscoring that finite-response CAPD is not claimed to dominate a fully known stationary-response method.

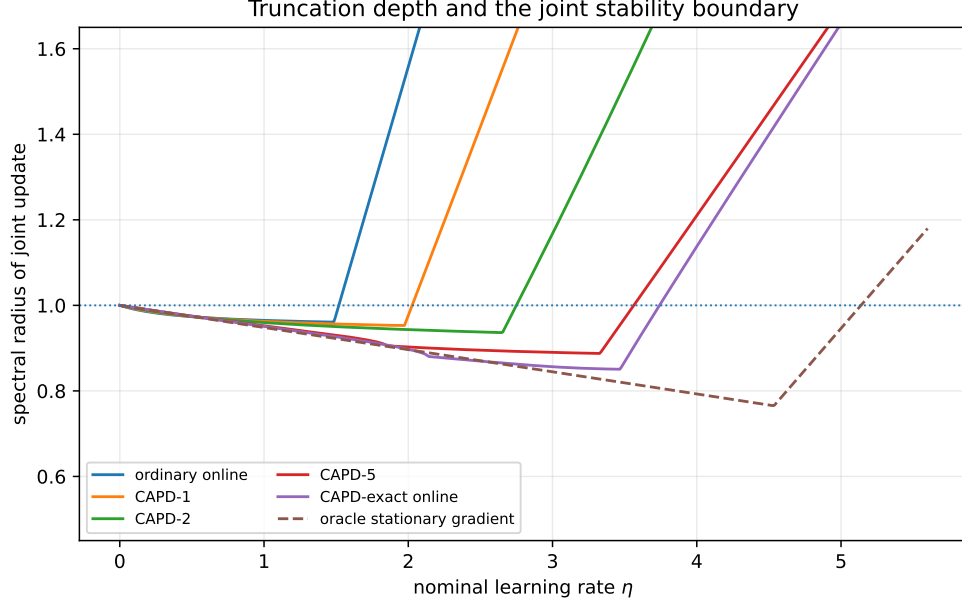


Figure 1: Spectral radius of the full joint update in the 3D quadratic benchmark. Response depth K changes the full Jacobian, and the stability boundary approaches that of the exact online correction. The oracle stationary-gradient baseline is shown separately.

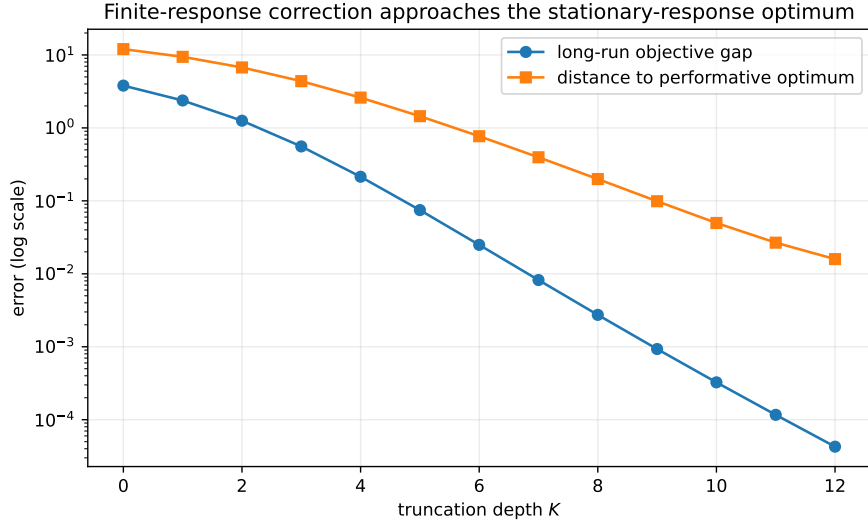


Figure 2: Geometric approach of finite-depth response corrections toward the stationary-response solution in the 3D quadratic benchmark.

7 Matched-Budget Comparison with Stateful PerfGD

The previous quadratic benchmark assumed that the response derivatives were available exactly. We now replace that oracle with the rolling finite-difference response estimator used by Stateful

PerfGD and compare the full stationary inverse with finite-depth propagation under matched response-estimation budgets. The environment and loss follow the authors’ released `ToyLinear.py` benchmark.¹

7.1 Stateful ToyLinear environment and shared estimator

Let $A_T \in \mathbb{R}^{d \times d}$ be symmetric negative definite and let $b \in \mathbb{R}^d$. The frozen-parameter stationary population mean and the one-step response are

$$\mu^*(\theta) = A_T \theta + b, \quad \mu_t = \delta \mu^*(\theta_t) + (1 - \delta) \mu_{t-1}. \quad (58)$$

The loss is

$$L(\theta, \mu) = -\theta^T \mu, \quad L^*(\theta) = -\theta^T (A_T \theta + b), \quad (59)$$

whose unique optimum is $\theta^* = (-2A_T)^{-1}b$.

Following Stateful PerfGD, define $\psi_t = (\theta_t, \hat{\mu}_{t-1})$ and use a rolling horizon of H observed response pairs. Centering the most recent sample gives matrices $\Delta\psi$ and $\Delta\hat{\mu}$, and the short-run Jacobian is estimated by

$$\hat{J}_m = \Delta\hat{\mu}(\Delta\psi)^\dagger = [\hat{D}_1, \hat{D}_2]. \quad (60)$$

This is the estimator in Algorithm 2 of Izzo, Zou, and Ying [3]. Stateful PerfGD then forms

$$\hat{S}_\infty = (I - \hat{D}_2)^{-1} \hat{D}_1, \quad (61)$$

whereas the matched CAPD- K variant uses

$$\hat{S}_K = \sum_{j=0}^{K-1} \hat{D}_2^j \hat{D}_1. \quad (62)$$

Both methods use the same estimated current mean and the same exploration perturbation in

$$\hat{g}_t = -\left(\hat{\mu}_t + \hat{S}^T \theta_t\right) + g_t, \quad \theta_{t+1} = \Pi_{[-R, R]^d}(\theta_t - \eta \hat{g}_t). \quad (63)$$

Thus the comparison changes only (61) versus (62); deployment, observation, Jacobian estimation, and model-update counts are identical.

7.2 Budget matching and protocol

We use $d = 5$, $R = 10$, $H = 15$, and 240 post-warmup deployments. Every method receives one population response and one noisy mean estimate per update, and common random numbers are used within each seed. Learning rate and exploration scale are selected from the same $8 \times 3 = 24$ candidate grid using five validation seeds; the reported values are then evaluated on 20 disjoint test seeds. We define the response time s so that, with θ frozen, the residual to $\mu^*(\theta)$ is 1% after s response steps:

$$\delta = 1 - 0.01^{1/s}. \quad (64)$$

We use $s \in \{16, 128, 512\}$. Mean-observation noise is either zero or Gaussian with standard deviation .03 in each coordinate.

¹Reference implementation: <https://github.com/zleizzo/gradual>. We reimplemented the benchmark from Algorithm 3 and the released ToyLinear environment so that both methods share exactly the same observations and random perturbations.

The stationary inverse is particularly demanding for slow response. Its exact gain is $1/\delta$, equal to 4.00 for $s = 16$, 28.30 for $s = 128$, and 111.68 for $s = 512$. In contrast, a fixed finite depth only propagates the estimated short-run response for K steps. In the clean case this creates deliberate bias: for $K = 5$, the fraction of the true ToyLinear stationary sensitivity retained is $1 - (1 - \delta)^5$, which falls from .763 at $s = 16$ to .165 at $s = 128$ and .044 at $s = 512$.

7.3 Results: clean accuracy versus noisy regularization

Table 4 reports the final long-run objective gap $L^*(\theta) - L^*(\theta^*)$. The comparison is intentionally two-sided. With clean response estimates, Stateful PerfGD’s full stationary sensitivity is essentially exact and dominates every finite truncation. Under slow noisy response, however, shallow truncations substantially outperform the full inverse on this benchmark. The effect is not due to additional data: all methods use the same $H = 15$ rolling response history and the same number of deployments.

Regime	Stateful PerfGD	CAPD-1	CAPD-2	CAPD-5
fast, clean ($s = 16$)	1.0×10^{-5}	4.930	2.880	0.341
slow, clean ($s = 128$)	2.4×10^{-5}	5.259	5.200	5.036
slow, noisy ($s = 128$)	61.98 ± 15.43	6.06 ± 0.40	7.34 ± 0.61	44.10 ± 7.51
very slow, noisy ($s = 512$)	66.97 ± 14.34	6.21 ± 0.55	5.16 ± 0.22	31.52 ± 10.20

Table 4: Stateful PerfGD versus CAPD- K on the ToyLinear stateful environment under identical response-estimation and deployment budgets. Entries are mean final long-run objective gap; noisy rows report mean $\pm$ standard error over 20 fresh test seeds. Hyperparameters for every method are selected from the same 24-candidate grid on five validation seeds. Clean estimates favor the full stationary inverse; slow noisy estimates favor shallow truncation.

At the benchmark dimension $d = 5$, a correction-only timing microbenchmark gives $7.05 \mu\text{s}$ for the Stateful PerfGD linear solve, compared with $1.57 \mu\text{s}$, $2.88 \mu\text{s}$, and $6.66 \mu\text{s}$ for CAPD-1, CAPD-2, and CAPD-5, respectively. Hence CAPD-5 is essentially compute-matched to the full solve, while the shallower corrections use less correction compute. The rolling pseudoinverse used to estimate (60) is shared by all methods and therefore does not favor CAPD.

Figure 3 summarizes the reversal between clean and noisy regimes. In the slow noisy case, Figure 4 shows the full learning trajectories. Stateful PerfGD remains the appropriate reference when the stationary response can be estimated accurately; finite depth becomes useful here specifically because noisy estimation of a slowly reacting population makes the stationary response sensitivity high variance.

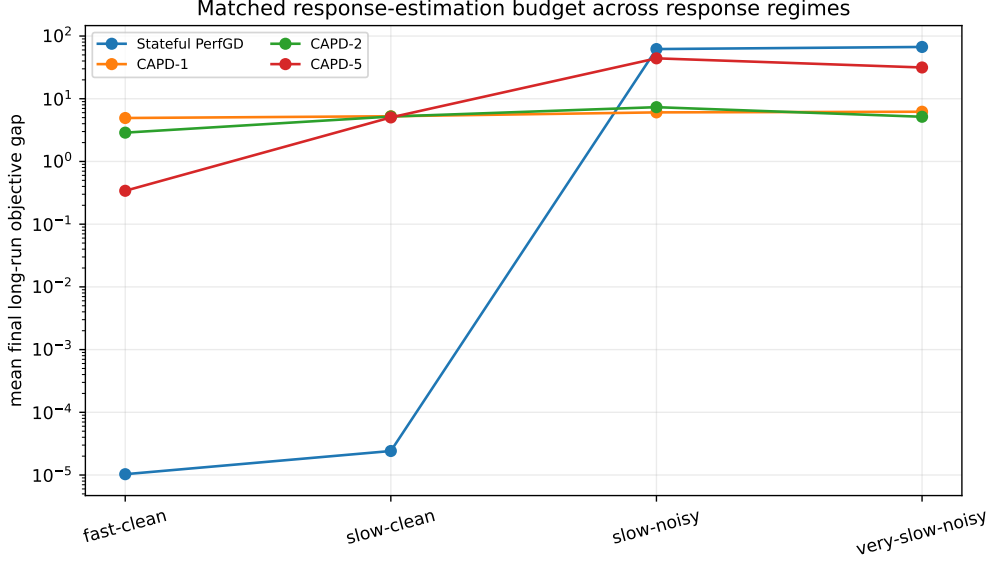


Figure 3: Final objective gap under matched response-estimation budgets. Full Stateful PerfGD dominates when response estimation is clean; shallow CAPD truncations are substantially more robust in the two noisy slow-response regimes.

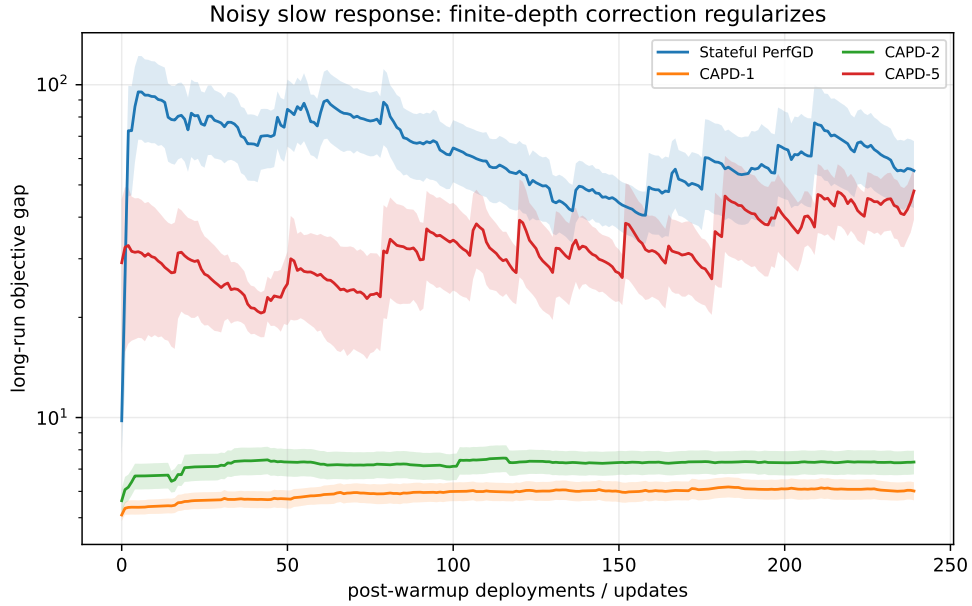


Figure 4: Twenty-seed mean trajectory in the slow noisy regime. All methods receive the same number of deployments and the same rolling response-estimation horizon. Shallow truncation prevents the large fluctuations produced by the estimated stationary inverse in this example.

The mechanism is consistent with Proposition 5.2. On the same test seeds, the mean Frobenius error $\|\hat{S} - S_\infty\|_F$ in the slow noisy regime is 12.76 for Stateful PerfGD, versus 5.17, 8.34, and 4.80 for CAPD-1, CAPD-2, and CAPD-5. The finite-depth methods therefore trade a known truncation

bias for substantially reduced response-estimation error. This is a regularization effect, not a claim that truncation is intrinsically more accurate than the stationary sensitivity.

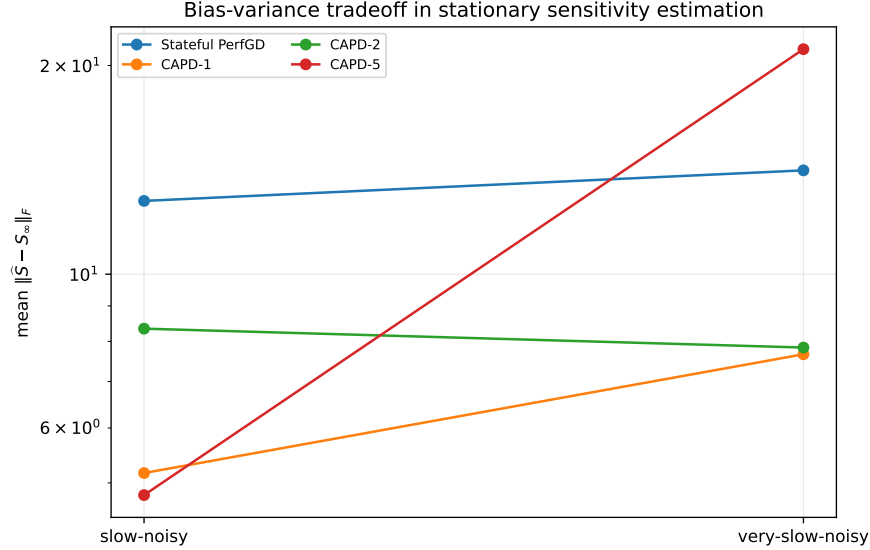


Figure 5: Total sensitivity error relative to the true stationary sensitivity in the noisy ToyLinear regimes. The full inverse and finite truncations react differently to response-estimation error; the best depth is problem dependent.

Interpretation. The experiment identifies a concrete regime in which finite response depth is not merely a cheaper approximation to Stateful PerfGD. It also regularizes an estimated near-stationary response. The clean rows show the cost of that regularization—potentially severe truncation bias for slow populations. The noisy rows show the opposite side of the tradeoff. Selecting K should therefore be viewed as choosing a response horizon, analogous to a bias-variance or model-horizon parameter, rather than as uniformly improving the stationary-response method.

8 Nonlinear Stateful Response under Matched Budgets

The ToyLinear comparison isolates the estimation mechanism, but its state-response map is affine. We next test whether the same finite-response regularization persists when the short-run environment is nonlinear in both the deployed model and the current population state. This is a stricter test because the rolling Jacobian estimator is now only locally valid and the stationary sensitivity varies along the trajectory.

8.1 Nonlinear environment and exact stationary reference

Let $\theta, \mu \in \mathbb{R}^d$ and define

$$\mu_{t+1} = (1 - \delta)\mu_t + \delta [A \tanh(\theta_t) + b + \rho \tanh(C\mu_t)], \quad (65)$$

where $\tanh$ acts componentwise. The frozen-parameter stationary state satisfies

$$\mu^*(\theta) = A \tanh(\theta) + b + \rho \tanh(C\mu^*(\theta)). \quad (66)$$

If

$$\rho \|C\|_2 < 1, \quad (67)$$

then the right-hand side of (66) is a contraction in μ , so $\mu^*(\theta)$ is unique. Writing

$$D_\theta = \text{diag}(\text{sech}^2(\theta)), \quad D_\mu = \text{diag}(\text{sech}^2(C\mu^*)), \quad (68)$$

implicit differentiation gives the exact stationary sensitivity

$$S_\infty(\theta) = \frac{d\mu^*}{d\theta} = (I - \rho D_\mu C)^{-1} A D_\theta. \quad (69)$$

The one-step response derivatives at a current state (θ, μ) are

$$D_1 m = \delta A \text{diag}(\text{sech}^2 \theta), \quad D_2 m = (1 - \delta)I + \delta \rho \text{diag}(\text{sech}^2(C\mu)) C. \quad (70)$$

At a stationary state, $(I - D_2 m)^{-1} D_1 m$ reduces exactly to (69).

We use the instantaneous loss

$$\ell(\theta, \mu) = \frac{\lambda}{2} \|\theta\|^2 - \theta^T \mu, \quad (71)$$

so the long-run performative objective is

$$L^*(\theta) = \ell(\theta, \mu^*(\theta)), \quad (72)$$

with exact gradient

$$\nabla L^*(\theta) = \lambda \theta - \mu^*(\theta) - S_\infty(\theta)^T \theta. \quad (73)$$

Equation (73) is used only to compute the ground-truth optimum and evaluation gap; neither competing online method receives the exact derivatives.

8.2 Protocol

We use $d = 3$, $\lambda = .8$, $\rho = .55$, and matrices chosen so that $\rho \|C\|_2 = .3025 < 1$. As in Section 7, both Stateful PerfGD and CAPD- K receive the same rolling response observations and use the same least-squares estimate

$$\hat{J}_m = [\hat{D}_1, \hat{D}_2]. \quad (74)$$

Stateful PerfGD uses $(I - \hat{D}_2)^{-1} \hat{D}_1$, while CAPD- K uses $\sum_{j=0}^{K-1} \hat{D}_2^j \hat{D}_1$. We use $H = 11$ response pairs, 220 post-warmup deployments, a common five-point learning-rate grid $\{.005, .01, .02, .04, .08\}$, and exploration scales $\{.01, .05\}$. Hyperparameters are selected on three validation seeds and the final numbers are evaluated on ten disjoint test seeds. The response time s again satisfies $\delta = 1 - .01^{1/s}$. We report a fast clean case ($s = 16$), a slow clean case ($s = 64$), and a slow noisy case ($s = 64$) with Gaussian mean-observation noise of standard deviation .02 per coordinate.

8.3 Results

Table 5 shows the same qualitative tradeoff as the linear benchmark, despite the fact that both $D_1 m$ and $D_2 m$ now vary along the trajectory. In the two clean regimes the full stationary correction is best, and increasing K moves finite-depth performance toward it. Under slow noisy response the ranking reverses: CAPD-1 and CAPD-2 have far smaller long-run objective gaps than Stateful PerfGD, whereas CAPD-5 is already sufficiently deep to re-amplify response-estimation error.

Regime	Stateful PerfGD	CAPD-1	CAPD-2	CAPD-5
fast, clean	0.00049 ± 0.00006	0.03492 ± 0.00056	0.02116 ± 0.00044	0.00531 ± 0.00022
slow, clean	0.01209 ± 0.00080	0.05207 ± 0.00089	0.04679 ± 0.00096	0.03745 ± 0.00080
slow, noisy	1.582 ± 1.186	0.0766 ± 0.0129	0.0555 ± 0.0122	0.424 ± 0.185

Table 5: Nonlinear stateful benchmark under matched deployment and response-estimation budgets. Entries are mean final long-run objective gap $L^*(\theta) - L^*(\theta^*) \pm$ standard error over ten fresh test seeds. The response is nonlinear in both θ and the current state μ .

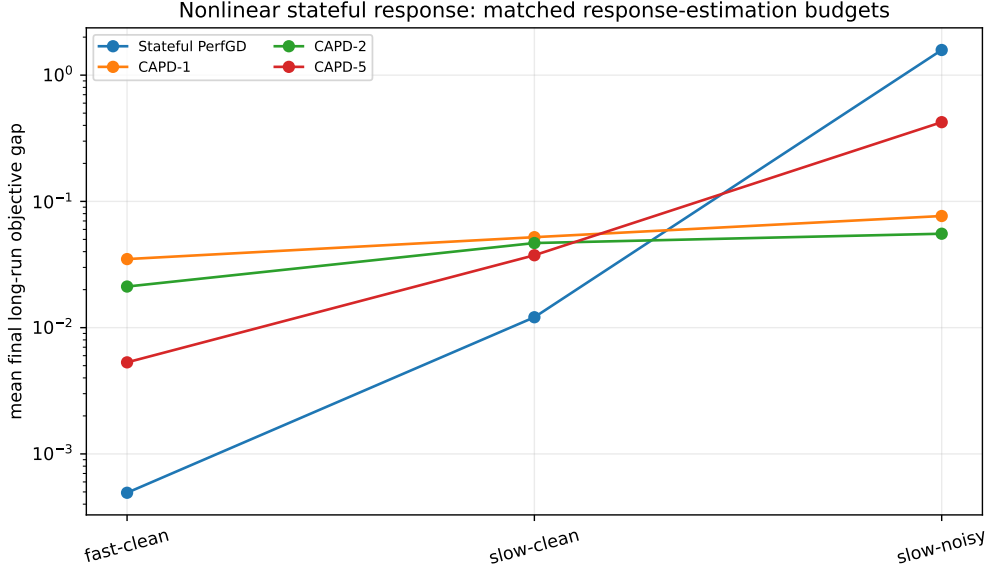


Figure 6: Matched-budget comparison on the nonlinear stateful response. Stateful PerfGD dominates under clean response estimation. With slow noisy response, finite-depth corrections are substantially more robust, and the best depth in this experiment is $K = 2$.

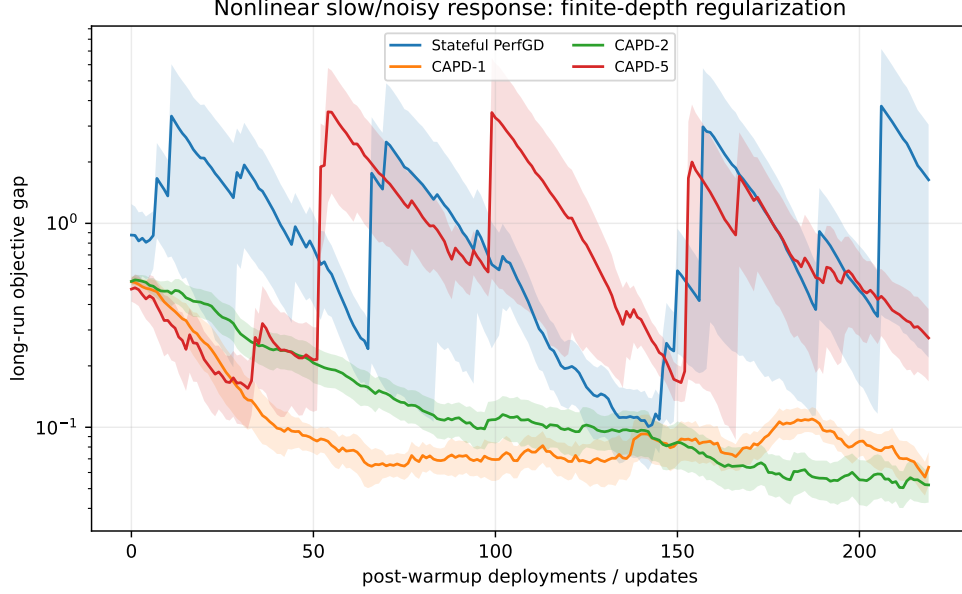


Figure 7: Ten-seed mean trajectory in the nonlinear slow/noisy regime. Stateful PerfGD and CAPD-5 exhibit large excursions, whereas CAPD-1 and CAPD-2 maintain smaller long-run objective gaps. Shaded bands are one standard error.

The sensitivity diagnostics make the depth effect visible. In the slow/noisy regime the mean total sensitivity errors $\|\hat{S} - S_\infty\|_F$ are 3.50, 2.15, 3.10, and 48.84 for Stateful PerfGD, CAPD-1, CAPD-2, and CAPD-5, respectively. The unusually large error at $K = 5$ comes from occasional multiplication of locally misestimated state Jacobians, illustrating that finite depth is not automatically stabilizing: it is a response-horizon parameter whose optimal value depends on both nonlinearity and observation quality.

Implication. The nonlinear experiment rules out the simplest explanation that the ToyLinear reversal is an artifact of affine response. It does not yet establish neural-network competitiveness, but it strengthens the interpretation of K as a bias–variance and response-horizon control: larger K reduces deterministic truncation bias in clean settings, while smaller K can suppress error propagation when a slowly reacting nonlinear environment is only locally estimated.

9 Adaptive Response Trust: Removing Manual Depth Selection

The stateful experiments show that the useful response depth depends strongly on response quality. Clean estimates favor deep propagation, whereas noisy slow response favors shallow truncation. This motivates an adaptive rule that controls response horizon and correction magnitude without preselecting one depth for every regime.

Let $g_{0,k}$ denote the direct or frozen-environment gradient at iteration k , and let

$$c_{K,k} = g_{K,k} - g_{0,k} \quad (75)$$

be the coupling correction produced by response depth K . Fix candidate depths

$$1 = K_1 < K_2 < \dots < K_J, \quad (76)$$

a trust parameter $0 < \tau_k < 1$, and a gate $0 < \gamma_{\min} < 1$. Adaptive CAPD chooses the deepest candidate satisfying

$$\|c_{K_j,k}\| \leq \tau_k \|g_{0,k}\|. \quad (77)$$

If no candidate satisfies (77), it starts from K_1 and uses

$$\gamma_k = \min \left\{ 1, \tau_k \frac{\|g_{0,k}\|}{\|c_{K_1,k}\| + \varepsilon} \right\}. \quad (78)$$

If $\gamma_k < \gamma_{\min}$, the response correction is gated off and the method takes the direct step. Otherwise the update is

$$\theta_{k+1} = \theta_k - \eta_k (g_{0,k} + \gamma_k c_{K_k,k}). \quad (79)$$

The gate is not a claim that direct training is globally optimal; it is a safety fallback when the estimated or exact endogenous correction is too large relative to the current direct gradient.

Algorithm 2 Adaptive Coupling-Aware Potential Descent (A-CAPD)

- 1: Compute direct gradient $g_{0,k}$.
 - 2: Build response corrections $c_{K_j,k}$ sequentially for $K_1 < \dots < K_J$.
 - 3: Select the largest K_j satisfying $\|c_{K_j,k}\| \leq \tau_k \|g_{0,k}\|$.
 - 4: **if** no candidate satisfies the trust condition **then**
 - 5: set $K_k = K_1$ and compute γ_k from (78);
 - 6: **if** $\gamma_k < \gamma_{\min}$ **then** set $K_k = 0$ and $\gamma_k = 0$;
 - 7: **end if**
 - 8: **else**
 - 9: set $\gamma_k = 1$.
 - 10: **end if**
 - 11: Update by (79).
-

The norm trust has a simple deterministic interpretation.

Theorem 9.1 (Frozen-potential descent under adaptive response trust). *Fix an environment state x and suppose $F(x, \cdot)$ has L -Lipschitz gradient. Let $g_0 = \nabla_{\theta} F(x, \theta)$ and let an adaptive response correction c satisfy*

$$\|c\| \leq \tau \|g_0\|, \quad 0 \leq \tau < 1. \quad (80)$$

For the update $\theta^+ = \theta - \eta(g_0 + c)$,

$$F(x, \theta^+) - F(x, \theta) \leq -\eta \left[(1 - \tau) - \frac{L\eta}{2} (1 + \tau)^2 \right] \|g_0\|^2. \quad (81)$$

Consequently every accepted adaptive correction is a strict descent direction for the frozen potential whenever

$$0 < \eta < \frac{2(1 - \tau)}{L(1 + \tau)^2}. \quad (82)$$

Proof. By Cauchy–Schwarz,

$$g_0^T (g_0 + c) \geq (1 - \tau) \|g_0\|^2, \quad \|g_0 + c\| \leq (1 + \tau) \|g_0\|. \quad (83)$$

The descent lemma gives

$$F(x, \theta^+) \leq F(x, \theta) - \eta g_0^T (g_0 + c) + \frac{L\eta^2}{2} \|g_0 + c\|^2, \quad (84)$$

which yields (81) and (82). $\square$

Theorem 9.1 is deliberately conditional on a frozen environment. Along the fully coupled trajectory, environment motion still contributes the injection term of (4); the theorem prevents the response correction itself from overturning the local descent direction, but it does not eliminate endogenous injection.

9.1 Residual-adaptive trust for estimated state response

When the short-run response Jacobian is estimated from rolling observations, the trust budget can also react to response-model uncertainty. Let $\hat{D}_k$ be the local least-squares response Jacobian fitted from centered response pairs $(\Delta U, \Delta Y)$, and define the normalized residual

$$r_k = \frac{\|\Delta Y - \hat{D}_k \Delta U\|_F}{\|\Delta Y\|_F + \varepsilon}. \quad (85)$$

In the stateful experiment below we use the fixed rule

$$\tau_k = \tau_{\min} + (\tau_{\max} - \tau_{\min})e^{-5r_k}, \quad (\tau_{\min}, \tau_{\max}) = (.25, .95). \quad (86)$$

Thus clean response estimation permits deeper correction, while noisy response fitting shrinks the trust region. The constants are not optimized separately by regime.

Using the nonlinear stateful benchmark of Section 8, candidate depths $\{1, 2, 5, 10\}$, and $\gamma_{\min} = .1$, A-CAPD uses mean depth 9.06 in the fast-clean regime and 7.62 in the slow-clean regime. In the slow-noisy regime, the normalized residual triggers the gate on 92.4% of updates and the mean effective depth falls to .16. The resulting long-run objective gaps are shown in Table 6. The adaptive rule does not beat the full stationary method in clean conditions and is not as accurate as the best hand-picked fixed depth in every regime; its role is to remove the need to know in advance whether the response estimate is clean or noise-amplified.

Regime	final objective gap	mean K	mean γ	fraction $K = 0$
fast, clean	.0278 $\pm$.0011	9.06	1.000	.000
slow, clean	.0467 $\pm$.0008	7.62	.999	.000
slow, noisy	.0587 $\pm$.0011	.15	.044	.924

Table 6: Residual-adaptive A-CAPD on the nonlinear stateful benchmark, ten fresh test seeds. Clean fits permit deep response propagation; noisy slow response causes the method to gate most response corrections off. Hyperparameters for the learner are selected from the same grids used in Section 8, while the trust rule (86) is fixed across regimes.

10 A Nonlinear Coupled Flip Transition

The previous section concerns linear stability. We now show explicitly what can appear after the coupled fixed point loses stability. Consider the scalar environment

$$x_{k+1} = ax_k + b\theta_k \quad (87)$$

and define $y = \theta - cx$. Let the local learning direction be

$$g(y) = y - \nu y^3 + \mu y^5, \quad \nu > 0, \quad \mu > 0, \quad (88)$$

which is the derivative of the bounded-below potential

$$f(y) = \frac{1}{2}y^2 - \frac{\nu}{4}y^4 + \frac{\mu}{6}y^6. \quad (89)$$

We study

$$\theta_{k+1} = \theta_k - \eta q g(\theta_k - cx_k), \quad (90)$$

where $q > 0$ represents a local response-correction scale. The frozen online gradient has $q = 1$; in the scalar linear-response example of the original draft, the local exact response correction has $q = \kappa = 1 - bc/(1 - a)$.

Assume

$$0 \leq a < 1, \quad b, c > 0, \quad bc < 1 - a. \quad (91)$$

Define

$$\eta_0 = \frac{2(1+a)}{1+a+bc}, \quad \eta_c = \frac{\eta_0}{q}, \quad s = \frac{1+a}{b}, \quad r = s + c. \quad (92)$$

Theorem 10.1 (Explicit coupled flip transition). *Under (91), the origin of (87)–(90) is locally asymptotically stable for $0 < \eta < \eta_c$ and loses stability at $\eta = \eta_c$ through a simple multiplier -1 ; the second multiplier remains strictly inside the unit disk and the crossing is transverse. For $\eta > \eta_c$ sufficiently close to η_c , there is a nonzero symmetric period-two orbit*

$$(x, \theta) = (x_\eta, -sx_\eta) \longleftrightarrow (-x_\eta, sx_\eta), \quad (93)$$

where

$$x_\eta^2 = \frac{\nu - \sqrt{\nu^2 - 4\mu(1 - \eta_c/\eta)}}{2\mu r^2}. \quad (94)$$

In particular,

$$|x_\eta| \sim \sqrt{\frac{\eta - \eta_c}{\nu r^2 \eta_c}} \quad \text{as } \eta \downarrow \eta_c. \quad (95)$$

Proof. Linearization at the origin gives

$$J_\eta = \begin{pmatrix} a & b \\ \eta qc & 1 - \eta q \end{pmatrix}. \quad (96)$$

The Jury conditions reduce to $0 < \eta q < \eta_0$, exactly as in the scalar linear model. At $\eta q = \eta_0$, the characteristic polynomial satisfies $p(-1) = 0$. Under (91) the second root is strictly inside the unit disk. Differentiating $p(\lambda, \eta q) = 0$ at $\lambda = -1$ shows the critical root has nonzero derivative, so the crossing is transverse.

For a symmetric two-cycle, the first equation in (93) gives $\theta = -sx$, hence $y = -rx$. The second map equation reduces, after dividing by nonzero x , to

$$1 - \nu r^2 x^2 + \mu r^4 x^4 = \frac{\eta_c}{\eta}. \quad (97)$$

Solving the resulting quadratic in x^2 and taking the small branch yields (94). Expanding the square root at $\eta = \eta_c$ gives (95). $\square$

Theorem 10.1 is a concrete dynamic-transition result: it identifies the critical learning rate and the invariant set that emerges immediately after loss of stability. More general coupled maps can be treated with center-manifold and flip/Neimark–Sacker normal forms [24]; the explicit model is included because every quantity is verifiable.

For $a = .5$, $b = .3$, $c = 1$, $\nu = 1$, and $\mu = .2$, one has $\eta_c = 5/3$ when $q = 1$. Numerical iteration reproduces the exact period-two amplitude to machine precision for $\eta = 1.68, 1.75, 1.90, 2.10, 2.50$. Figure 8 shows the bifurcation diagram and later post-critical complexity.

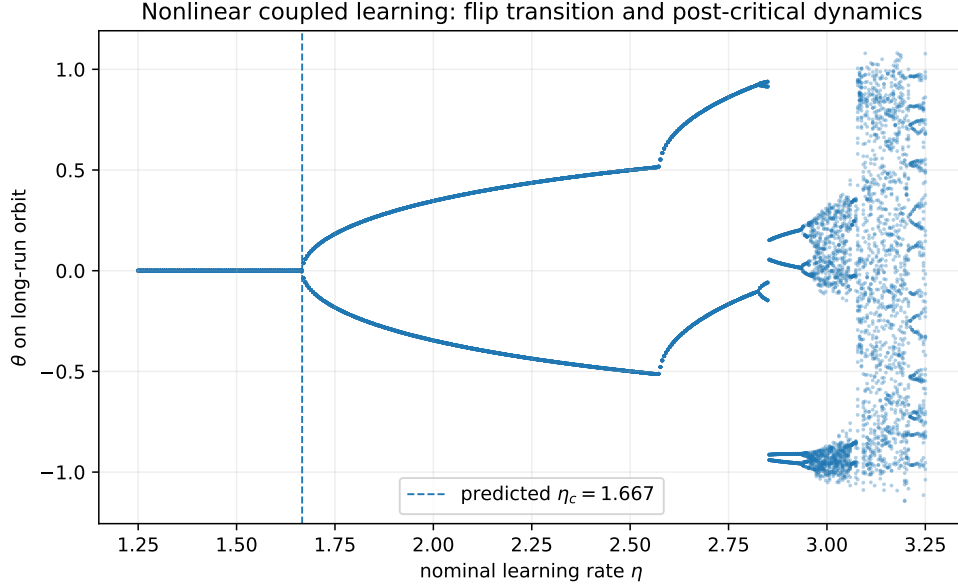


Figure 8: Bifurcation diagram for the nonlinear coupled model with $q = 1$. The vertical line is the analytically predicted flip threshold $\eta_c = 5/3$. A period-two branch appears immediately beyond the threshold; additional dynamics arise for larger learning rates.

11 Scope, Limitations, and Discussion

The results support a narrow conclusion: finite response depth can be useful when the stationary response is noisy, expensive, or dynamically incomplete. They do not establish general superiority over Stateful PerfGD, plug-in performative optimization, Markovian methods, or regularized stationary-response estimators. In fact, the clean-response experiments show the opposite: when the stationary sensitivity is accurately estimated, full response is preferable.

The regularized stationary-response audit shows that finite response is one regularization mechanism among several: Tikhonov and spectral clipping remove much of the instability of the raw stationary inverse. Theorem 5.4 sharpens the finite-horizon case by separating deterministic truncation bias from stochastic Jacobian-estimation error and by defining an oracle preferred depth. Section 5.1 replaces the earlier OLS standard-error proxy by a finite-sample uncertainty theorem for the actual rolling reference-difference regression. The bound is pathwise in the adaptive trajectory and accounts explicitly for measurement error in both the response and the lagged-state covariate. The covariance-aware refinement represents repeated-reference differences through incidence matrices and self-normalizes them by the observed pseudoinverse geometry; this shrinks the noisy ToyLinear radii by about 4.6 relative to the coordinatewise worst-case treatment. Its certified selector still recovers the best evaluated depth in three of four regimes but remains too conservative to distinguish $K = 1$ from $K = 2$ in the very-slow/noisy case. The direction-aware ellipsoid in Theorem 5.10 reduces this scalar conservatism and improves the hardest ToyLinear regime, but its occasional deep selections hurt the ordinary slow/noisy case. Sequentially uniform uncertainty and oracle

inequalities linking sensitivity error to online objective loss remain open.

The transition theorem is deliberately low dimensional. General high-dimensional transitions require center-manifold or spectral-projection arguments near critical multipliers. Adaptive CAPD also uses hand-set trust constants and candidate depths. Its deterministic guarantee controls only the frozen potential; environmental motion can still inject loss into the coupled trajectory.

12 Conclusion

Learning under endogenous feedback is not characterized only by an eventual stationary response. Between frozen-environment training and full stationary-response optimization lies a finite-response regime whose depth controls both approximation bias and sensitivity to response-estimation error. The stochastic response-depth theorem makes this tradeoff quantitative: geometric bias decreases with K , while Jacobian-estimation error accumulates with the propagated horizon, producing an interior preferred depth in slow noisy regimes and explicit approximations for $K^*(q, \sigma, n)$. A covariance-aware confidence-bound selector based on the same rolling response regression carries the exact shared-reference differencing geometry through the observed pseudoinverse, reducing the noisy-regime Jacobian radii by about 4.6 relative to a coordinatewise certificate. A direction-aware ellipsoid then propagates uncertainty only through the realized finite-response directions, improving the very-slow/noisy selector while exposing that local sensitivity certification alone does not guarantee the best online depth. It recovers the best evaluated depth in three of four ToyLinear regimes and defaults conservatively to $K = 1$ in the hardest very-slow/noisy case, where the fixed-depth oracle is $K = 2$. The experiments show that this regime is consequential: response depth changes the full learner–environment Jacobian and remains competitive with explicitly regularized stationary-response methods in the most difficult noisy tests. The stationary hypergradient remains the correct endpoint when it can be estimated accurately; finite response provides a controlled transient alternative when that endpoint is statistically fragile or dynamically inappropriate.

A Empirical Check of the Coupling Diagnostic

Theorem 3.3 predicts that the first-order balance $C - D$ should identify the sign of a finite-step potential change when the step is not too large relative to the local curvature margin. We test this on a three-dimensional quadratic potential with deliberately strong moving-target injection. For 10^4 random states at each step size, the environment is advanced by an Euler step and the learner by a simultaneous gradient step. Table 7 reports the fraction of trials in which the threshold $\chi_+ > 1$ correctly predicts an increase in the realized potential.

Euler step h	sign accuracy	fraction of realized increases
0.025	0.9941	0.1335
0.050	0.9865	0.1334
0.100	0.9721	0.1517
0.200	0.9420	0.1872

Table 7: Finite-step diagnostic benchmark. Accuracy decreases as the Taylor remainder grows, consistent with Theorem 3.3. The test is diagnostic rather than a claim about a neural optimizer.

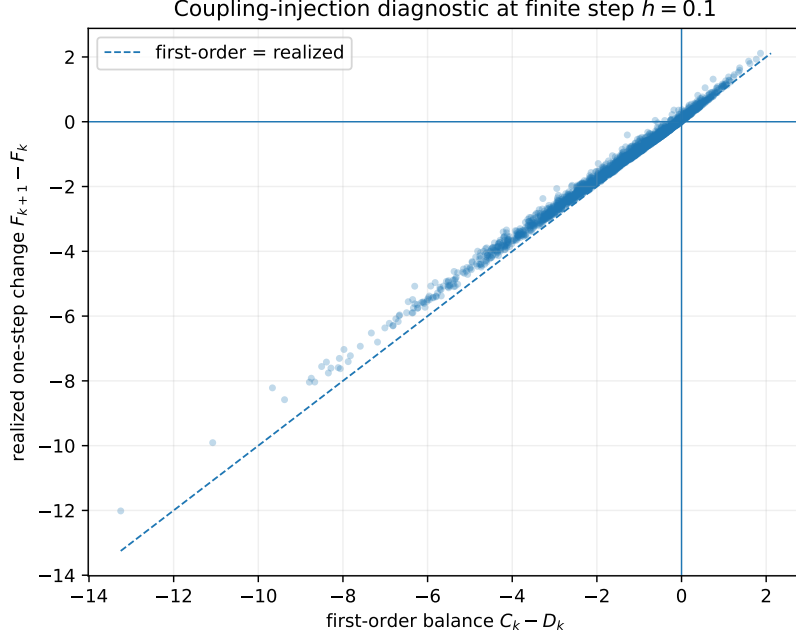


Figure 9: First-order potential balance $C_k - D_k$ versus the exact realized one-step change at $h = .1$. The sign agreement is 97.21% over 10^4 random states.

The revised use of χ_+ is therefore deliberately limited: it is a potential-balance certificate and monitoring statistic. Whether it should control a practical learning-rate scheduler is left to future empirical work rather than asserted here.

B Additional Linear-Algebra Details

For the quadratic benchmark, the K -step direction can be written without explicitly storing adjoint iterates. Since

$$v_K = - \sum_{j=0}^{K-1} (A^T)^j C^T (R\theta - Cx - d), \quad (98)$$

one obtains

$$g_K = (R - CP_K)^T (R\theta - Cx - d) + \lambda\theta = D_K^T (R\theta - Cx - d) + \lambda\theta. \quad (99)$$

Thus the joint Jacobian is

$$\mathcal{J}_{K,\eta} = \begin{pmatrix} A & B \\ \eta D_K^T C & I - \eta(D_K^T R + \lambda I) \end{pmatrix}. \quad (100)$$

All stability thresholds reported in Table 3 are obtained directly from $\rho(\mathcal{J}_{K,\eta}) = 1$ on a dense one-dimensional grid and can be reproduced by the accompanying script.

C Reproducibility

The analytic verification and quadratic stability calculations are reproduced by `code/benchmarks.py`. The matched-budget Stateful PerfGD comparison is reproduced by `code/stateful_perfgd_vs_`

capd.py and code/evaluate_matched_budget.py. The nonlinear stateful experiment is reproduced by code/nonlinear_stateful_compare.py. Adaptive stateful response trust is reproduced by code/adaptive_capd_stateful.py. The nonlinear period-two comparison uses the exact branch formula in Section 10.

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